

5.5 Connecting Ratios and Rates

Lesson Introduction

 Benchmark	MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.		 Learning Target	I can make connections between ratios, rates, and unit rates.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.		N/A	
 Essential Questions	- What strategies can you use to calculate the unit rate and associated ratios? - What two mathematical operations are used to convert measurement in one unit to measurement in another unit? - If work is being done at a constant rate by one person, and a different constant rate by another person, how do you convert them to compare them directly?			
 Lesson Narrative	Students explore how to scale recipes up and down using ratios, multiplication, and division, connecting ratios more formally to unit rates. Students then build on this understanding as they use ratios and rates to describe relationships in the context of speed.			
 Component Summary	5.5.1 Warm-Up (~5 min.)	Analyze examples of mathematics in nature (<i>SE p. 193</i>)	5.5.4 Try It (10 min.)	Practice comparing the ratio, rate, and unit rates of a slug and a snail (<i>SE p. 196</i>)
	5.5.2 Guided Instruction (10 min.)	Solve ratio and rate problems using double number lines and ratio tables (<i>SE p. 194</i>)	5.5.5 Your Turn! (10 min.)	Practice solving real-world problems involving ratios, rates, and unit rates using a double number line and a ratio table (<i>SE p. 197</i>)
	5.5.3 Collaboration (10 min.)	Compare the ratio, rate, and unit rates of two runners (<i>SE p. 195</i>)	5.5.6 Wrap-Up (~5 min.)	Student reflection on which topics they are still unsure and plan to improve understanding of these topics (<i>SE p. 197</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Markers	
			<u>Additional Preparation</u> Warm-Up 5.5.1 contains a video to accompany the question prompts.	

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may think that the object that traveled the farthest is the fastest (if they do not consider how much time it took). - Students may struggle to make connections between tape diagrams, double number lines, and ratio tables, believing them to be different quantities rather than the same quantity represented in different ways.		- Remind students that a rate is a comparison of two quantities that have different units of measure and that the ratio is not complete unless it has units. - Consider displaying the ratio table horizontally to help students connect it to a tape diagram or double number line. - Create an anchor chart and allow students to create reference cards with instructions and examples to aid them in applying ratio relationships to solve real-world problems.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Think-Pair-Share Consider structuring 5.5.3 Collaboration with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.		MA.K12.MTR.2.1: Multiple Representations Students apply ratio relationships to solve real-world problems. Across the lesson components, the ratios are represented in ratio tables and double number lines. Students make connections among these representations and determine the most efficient representation to use and how to use it to solve problems.	
	Consider using double number lines and equivalent ratio tables to help students visualize the ratios in the 5.5.4 Try It and 5.5.5 Your Turn! components of the lesson. Consider having students represent the problem using both representations as an additional scaffold, if needed.			
Lesson Notes:				

5.5.1 Warm-Up: Why do Honeybees Love Hexagons?

Component Overview: Students justify why a hexagonal shape allows bees to store the most honey using the least amount of wax and share another example of mathematics in nature.



5.5.1 Warm-Up: Why do Honeybees Love Hexagons?

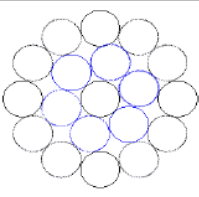
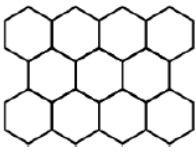
Honeybees are actually amazing mathematicians. Scientists claim they can even calculate angles. They also live inside one of the most mathematically efficient architectural designs there is: the beehive. It takes a lot of work for bees to produce wax. The hexagonal shape of the cells in a honeycomb allows bees to store the most amount of honey while using the least amount of wax. This design likely evolved over time.



Consider allowing students to discuss question prompts rather than writing their answers for classes that engage better using routines like Turn and Talk or Think-Pair-Share.

Use sentence stems to encourage conversation such as: “From least to greatest, the design with the greatest area (most honey) is...” and “From least to greatest, the design with the least perimeter (least wax) is ...” “Therefore, the best design (most honey and least wax) is...”

Consider the shapes used in the designs shown.

Triangular Cells	Circular Cells	Hexagonal Cells
		

- How do the designs demonstrate that the hexagonal shape allows bees to store the most amount of honey using the least amount of wax?
Answers will vary. The hexagonal shape allows bees to store the most amount of honey using the least amount of wax because there are no spaces in the pattern which allow for no wasted space as is the case of the circular shape.
- This is an example of mathematics in nature. What is another example of mathematics in nature you can think of?
Answers will vary. Repeated patterns on leaves are an example of mathematics in nature.

5.5.2 Guided Instruction: Connecting Ratios and Rates

Component Overview: Students use a double number line and corresponding table to represent a ratio used to make a drink mixture, then use what they know about unit rate to determine which operations to use when scaling the recipe up and down. Engage the class in a discussion to highlight questions 3-5 and solidify understanding for how to find equivalent ratios.

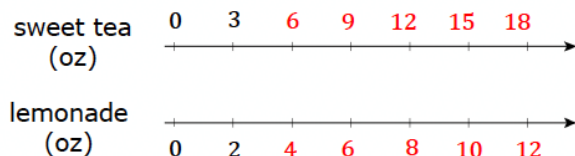


5.5.2 Guided Instruction: Connecting Ratios and Rates

A given ratio or rate can be used to generate many equivalent ratios, which can be helpful in solving problems. Double number lines, tape diagrams, mixture charts, and ratio tables are useful tools for finding equivalent ratios.

- Noah's favorite drink is a mixture of sweet tea and lemonade. He uses the ratio of 3 ounces (oz) of sweet tea to 2 oz of lemonade to make it.

A double number line diagram is shown to represent the situation. Complete the double number line diagram.



A table can also be used to represent this relationship.

- Complete the table to show how many ounces of each beverage are needed to make different-sized batches of Noah's favorite drink.

Sweet Tea (oz)	Lemonade (oz)
3	2
21	14
15	10
4.5	3
48	32
1.5	1

Recipes are scaled up to make larger batches with the same taste. They can also be scaled down to make smaller batches with the same taste. Equivalent ratios can be used to determine the amount of ingredients needed when scaling a recipe up or down.



This Guided Instruction guides students in how to apply the knowledge they have learned in this unit to connect the double number line and table representations. Students may not need this to be modeled, as it is practice from skills they have been building in previous lessons.

Bring the class together to complete questions 3-5 to highlight the ways in which equivalent ratios are created before students continue working with ratios and unit rates in tables. Ensure students can articulate the ways in which equivalent ratios can be determined.

5.5.2 Guided Instruction: Connecting Ratios and Rates (continued)

3. When scaling a recipe up, such as to find how many ounces of sweet tea to mix with 32 ounces of lemonade, which operation is used?

Multiplication

4. When scaling a recipe down, such as to find how many ounces of lemonade to mix with 1.5 ounces of sweet tea, which operation is used?

Division

5. William noticed he could use similar thinking with the double number line diagram. He drew an arrow from 3 to 15 on the top number line and another arrow from 2 to 10 on the bottom number line. Complete the statement to describe what he saw.

William noticed the ratio $\frac{3}{2}$ can be used to find the ratio

$\frac{15}{10}$ by

- adding 12 to both parts of the ratio.
- multiplying both parts of the ratio by 5.
- dividing both parts of the ratio by 5.

Equivalent ratios can be found by multiplying or dividing each part of a ratio by the same value.

6. Show how the ratios 3:2 and 48:32 from the table are equivalent, using either multiplication or division.

$$\frac{3}{2} \times \frac{16}{16} = \frac{48}{32} \text{ or } \frac{48}{32} \div \frac{16}{16} = \frac{3}{2}$$

7. Shay noticed the table of Noah's favorite drink included a unit rate. Circle the unit rate in the table. Then, complete the statement to explain its meaning in context.

Shown in table.

This unit rate means there are 1.5 ounces of sweet tea in the drink for each ounce of lemonade.



Students may confuse when to multiply to find the equivalent ratio and when to divide, or they may not understand the vocabulary of “scaling up” and “scaling down.” Have students consider what is happening when they are creating more/less of a recipe, and what operation can help them accomplish that. It will be helpful to connect this to a visual for students who may struggle.

In the discussion, guide students to make connections between “equal groups” and multiplication and division.

- *What is happening when you scale up a recipe?*
- *Will there be more or less of the drink?*
- *What operation can be used to create more?*
- *What happens when you scale down a recipe?*
- *What operation can be used to create less?*
- *How can you show this using a double number line?*

5.5.3 Collaboration: Considering Ratios and Rates in the Context of Speed

Component Overview: Students work with a partner to compare the ratio, rate, and unit rate of two runners. Consider structuring this section with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning.



5.5.3 Collaboration: Considering Ratios and Rates in the Context of Speed

Mai and Jada each ran on a treadmill. The treadmill display shows the distance, in miles, each person ran and the amount of time it took them, in minutes and seconds.

Mai's Treadmill Display	Jada's Treadmill Display

- Discuss with your partner the similarities and differences between Mai's and Jada's workouts. Write them in the table below.

Answers will vary.

Similarities	Differences
<i>Distance = 3.0 miles</i> <i>Incline = 10</i> <i>Level = 12</i> <i>Pulse = 125</i>	<i>Minutes = 24:00 and 30:00</i> <i>Calories = 481 and 411</i> <i>Pace = 8:00 and 10:00</i>

- Express each person's workout as a ratio of distance to time, including units.

	Ratio of distance to time
Mai's Workout	<i>3 miles : 24 minutes</i>
Jada's Workout	<i>3 miles : 30 minutes</i>



Think-Pair-Share

Consider structuring questions 1-3 with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.

5.5.3 Collaboration: Considering Ratios and Rates in the Context of Speed (continued)

3. Each person ran at a constant rate by setting the speed on their treadmill. Discuss with your partner who ran at a faster rate using the ratios.

The ratio of distance to time for Mai is 3 : 24. The ratio of distance to time for Jada is 3 : 30. It took Jada 30 minutes to run 3 miles but only took Mai 24 minutes to run 3 miles. Therefore, Mai ran faster.

4. Eli thought it would be helpful to rewrite the ratios as unit rates when deciding who ran faster. His incomplete work is shown. Work with your partner to complete his work.

Mai's Workout	Jada's Workout
$\div 3 \rightarrow 24 \text{ min} : 3 \text{ mi} \rightarrow \div 3$ $\div 8 \rightarrow 8 \text{ min} : 1 \text{ mi} \rightarrow \div 8$ $\div 8 \rightarrow 1 \text{ min} : \frac{1}{8} \text{ mi}$	$\div 3 \rightarrow 30 \text{ min} : 3 \text{ mi} \rightarrow \div 3$ $\div 10 \rightarrow 10 \text{ min} : 1 \text{ mi} \rightarrow \div 10$ $\div 10 \rightarrow 1 \text{ min} : \frac{1}{10} \text{ mi}$

5. Complete the statements.

It took Mai 8 minutes to run 1 mile, and it took Jada 10 minutes to run 1 mile. This shows that ☐ Mai ☐ Jada ran faster because it took her ☐ less ☐ more time to run the same distance.

In 1 minute, Mai ran $\frac{1}{8}$ mile and Jada ran $\frac{1}{10}$ mile. This shows that ☐ Mai ☐ Jada ran faster because she ran ☐ less ☐ more distance in the same amount of time.



Students may not understand that a faster time means Mai ran at a faster rate, instead thinking that running for 30 minutes means Jada ran faster or more miles because 30 is more than 24.

Make sure to address this misconception if you see it arising.

- *Who ran for longer? How do you know?*
- *Did both students run the same number of miles?*
- *What does it mean if Mai ran the same number of miles in less time?*
- *Is she running faster or slower than Jada? Is $\frac{1}{3}$ more or less than $\frac{1}{10}$?*



The table provides a visual entry point. Students can use different colors to keep track of their work and see who ran faster. Encourage partners to discuss their work in the table to provide an auditory entry point.

5.5.4 Try It: Rates and Ratios in Context of Speed

Component Overview: Students work with a partner to determine whether the slug or snail is moving faster based on the ratio of centimeters traveled to seconds. Students evaluate a statement and explain if they agree or disagree based on the ratios.



5.5.4 Try It: Rates and Ratios in Context of Speed

1. A slug travels 3 centimeters in 3 seconds. A snail travels 6 centimeters in 6 seconds. Ethan says, "The snail was traveling faster because it went a greater distance."
 - a. Discuss Ethan's statement with your partner.
Answers will vary. Some may agree or disagree depending on their understanding of unit rates.
 - b. Explain whether you agree with Ethan using ratios or rates.
I disagree with Ethan. The slug's unit rate is $\frac{3}{3} = 1$ cm per second. The snail's unit rate is $\frac{6}{6} = 1$ cm per second. They both have the same unit rate.



Encourage students to depict the situation using a double number line, table, or other visual representation. Highlight any creative representations as a class.



Students should be applying what they have already learned about ratios and unit rates to their discussions about Ethan's statement. Encourage students to use the formal vocabulary of ratios, rates, and unit rates in their discourse and explanation. Students should be evaluating the rate of the slug and snail to prove their explanations.

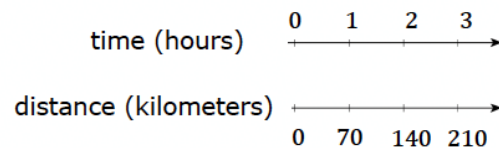
5.5.5 Your Turn!

Component Overview: Students work independently to solve real-world problems involving ratios, rates, and unit rates using a double number line and a ratio table. Pose questions to check for understanding and help students consolidate their understanding after students have completed both questions.



5.5.5 Your Turn!

- Nadia is driving at a constant speed on a long stretch of highway on a long drive from Melbourne, Florida, to Atlanta, Georgia. Her distance traveled over time is shown on the double number line.



How far did Nadia travel at this speed for 5 hours?
Explain or show your reasoning.

*The unit rate of Nadia's travel is 70 kilometers per hour.
Nadia traveled $70 \times 5 = 350$ kilometers.*

- Two cleaning solutions were created to clean tile floors as shown.

Mixture A	Mixture B
5 cups of water 2 cups of vinegar	15 cups of water 8 cups of vinegar

- Explain why the mixtures will not result in the same cleaning solution.

The ratio of water to vinegar for Mixture A is $\frac{5}{2} = 2.5$. This is a 2.5:1 ratio. The ratio of water to vinegar for Mixture B is $\frac{15}{8} = 1.875$. This is a 1.875:1 ratio. Mixture A has a greater ratio of water to vinegar which results in a different cleaning solution.

- Explain which solution will have a stronger smell of vinegar.

Mixture B has 1.875 cups of water per 1 cup of vinegar. Mixture A has 2.5 cups of water per 1 cup of vinegar. Therefore, Mixture B would have a stronger smell because it would be less concentrated than mixture A.



Students may struggle to explain why the mixtures will not result in the same cleaning solution. Encourage students to consider what they know about ratios and unit rates to help explain their answer.

How can you determine ratio of water to vinegar for each solution? (Compare how many cups for water and how many for vinegar)

So how could you represent this comparison visually or using math? (Double number line, table, or ratio)

Do you think the mixture with more or less vinegar will smell more strongly? Why?

5.5.6 Wrap-Up: Growth Mindset

Component Overview: Students reflect on which topics they are still unsure about and develop a plan to improve their understanding of these topics. This activity helps students enhance their metacognitive skills and encourages them to take ownership of their learning.



5.5.6 Wrap-Up: Growth Mindset

Complete each statement.

1. Today I am still unsure about ...

Answers will vary. Today I am still unsure about the difference between ratios and rates.

2. To fill this gap, I intend to ...

Answers will vary. To fill this gap, I intend to study my notes.



The 5.5.6 Wrap-Up asks students to write an explanation of which topics they are still unsure about. Students who struggle with written expression could rate each problem from the lesson using a coding system such as:

- 0 = don't understand
- 1 = partially understand
- 2 = completely understand